

ADR-0001: The LLM orchestrates and narrates; it never calculates

Status: Accepted **Date:** 2026-06-01 **Deciders:** Product owner, Eng lead

Context

This is a decision-support tool for CxOs. Numbers (price/sqft, amenity premium, variance, absorption) drive capital-allocation and pricing decisions. A hallucinated figure that looks plausible is a catastrophic, trust-destroying failure. LLMs are strong at language and orchestration but unreliable at arithmetic and prone to inventing confident numbers. Credible source data is also sparse, so the provenance of every figure must be explicit.

Decision

The LLM (Gemini via LangChain4j) **only** decides which tool to call, with what parameters, and phrases the answer. **All numbers are produced by deterministic, unit-tested Java services** and passed to the LLM as tool results carrying a value, a credibility tier, and a source reference. The LLM may not perform arithmetic, estimate, average, or infer a number that no tool returned. If a tool returns no data, the agent says "no data" — it never fills the gap with an estimate. RAG supplies qualitative context only, never figures.

Options Considered

Option A: LLM computes / free-form reasoning over raw data

Dimension	Assessment
Complexity	Low (less plumbing)
Cost	Low upfront
Reliability	Unacceptable — hallucinated/incorrect numbers
Auditability	Poor — no traceable derivation

Pros: Fast to build; flexible. **Cons:** Non-deterministic numbers; no audit trail; fatal for a CxO tool.

Option B: Deterministic tools; LLM orchestrates + narrates (chosen)

Dimension	Assessment
Complexity	Medium — fixed tool layer over tested services
Cost	Medium
Reliability	High — every figure deterministic and reproducible
Auditability	High — each number cites its source observation(s)

Pros: Trustworthy, testable, auditable; the math can be verified independently of the model. **Cons:** More engineering; the agent can only answer what a tool exposes (acceptable, even desirable).

Trade-off Analysis

We trade some flexibility and upfront speed for correctness, reproducibility, and auditability. For a tool whose entire value is trustworthy strategic numbers, that trade is non-negotiable.

Consequences

- **Easier:** unit-testing analytics; citing every figure; swapping the LLM provider.
- **Harder:** every new quantitative capability needs a real, tested tool — no shortcuts via prompting.
- **Revisit:** never for correctness-critical numbers. Qualitative summarization latitude can grow.

Action Items

1. ☐ Expose analytics/reconciliation only through a fixed, audited `@Tool` set.
2. ☐ System prompt forbids unsourced figures; add a response check that each number maps to a tool output.
3. ☐ Tool results carry `{value, tier, source_ref, computed_at}`.